

List of graphs and charts used in Microsoft Power BI:

- ❖ Bar Chart
- ❖ Column Chart
- ❖ Stacked Bar Chart
- ❖ Stacked Column Chart
- ❖ Clustered Bar Chart
- ❖ Clustered Column Chart
- ❖ Line Chart
- ❖ Area Chart
- ❖ Stacked Area Chart
- ❖ Combo Chart (Line and Column)
- ❖ Pie Chart
- ❖ Donut Chart
- ❖ Funnel Chart
- ❖ Waterfall Chart
- ❖ Gauge Chart
- ❖ KPI Visual
- ❖ Card (Single Number)
- ❖ Multi-row Card
- ❖ Table
- ❖ Matrix
- ❖ Treemap
- ❖ Scatter Chart
- ❖ Bubble Chart
- ❖ Filled Map (Choropleth)
- ❖ Map (Basic Map)
- ❖ ArcGIS Maps for Power BI
- ❖ Shape Map

- ❖ Decomposition Tree
- ❖ Ribbon Chart
- ❖ Histogram
- ❖ Scatter with Play Axis (Animated)
- ❖ Q&A Visual
- ❖ Smart Narrative
- ❖ R Visual
- ❖ Python Visual

1. Comparison Charts

Used to compare values across categories.

- ❖ Bar Chart
- ❖ Column Chart
- ❖ Clustered Bar Chart
- ❖ Clustered Column Chart
- ❖ Stacked Bar Chart
- ❖ Stacked Column Chart
- ❖ Ribbon Chart
- ❖ Combo Chart (Line and Column)

2. Trend / Time-Series Charts

Show changes over time or patterns.

- ❖ Line Chart
- ❖ Area Chart
- ❖ Stacked Area Chart
- ❖ Combo Chart (Line and Column)
- ❖ Scatter Chart with Play Axis (for time animation)

3. Composition Charts

Show how a whole is divided into parts.

- ❖ Pie Chart
- ❖ Donut Chart
- ❖ Funnel Chart
- ❖ Treemap
- ❖ Stacked Column Chart
- ❖ Stacked Bar Chart
- ❖ Waterfall Chart

4. Distribution Charts

Show spread, range, and frequency of data.

- ❖ Scatter Chart
- ❖ Bubble Chart
- ❖ Histogram
- ❖ Box and Whisker Chart (*via custom visual*)

5. KPI & Summary Visuals

Highlight key metrics and status.

- ❖ KPI Visual
- ❖ Card (Single Number)
- ❖ Multi-row Card
- ❖ Gauge Chart
- ❖ Smart Narrative

6. Table-Based Visuals

Show detailed data with rows and columns.

- ❖ Table
- ❖ Matrix

7. Geographic / Map Visuals

Plot data geographically.

- ❖ Map (Basic Map)
- ❖ Filled Map (Choropleth)
- ❖ ArcGIS Maps for Power BI
- ❖ Shape Map

8. Advanced / Analytical Visuals

Enable deeper analysis or AI-driven insights.

- ❖ Decomposition Tree
- ❖ Q&A Visual
- ❖ R Visual
- ❖ Python Visual

1.Comparison Charts

Purpose: Compare categories side-by-side.

Icon	Visual Name	Use Case	When to Use
	Bar Chart	Compare sales by region	When categories have long names (horizontal bars).
	Column Chart	Compare revenue by product	When showing differences between categories vertically.
	Clustered Bar / Column Chart	Compare multiple series for same categories	When you need side-by-side bars.
	Stacked Bar / Column Chart	Show contribution of each part to total	When you want to see overall size and part proportions.
	Ribbon Chart	Show category rank changes over time	Ideal for leaderboard/rank shift visuals.

Icon	Visual Name	Use Case	When to Use
	Combo Chart (Line & Column)	Compare trends and totals together	When you need to overlay a trend with bars.

2. Trend / Time-Series Charts

Purpose: Show changes over time.

Icon	Visual Name	Use Case	When to Use
	Line Chart	Track monthly revenue	When you want continuous time-based trends.
	Area Chart	Show cumulative trend	When you want to emphasize total volume.
	Stacked Area Chart	Show trend contribution by category	When comparing category growth over time.
	Scatter Chart with Play Axis	Animated trend movement	When showing how metrics evolve over time.

3. Composition Charts

Purpose: Show parts of a whole.

Icon	Visual Name	Use Case	When to Use
	Pie Chart	Show market share	When you have ≤ 5 categories.
	Donut Chart	Show composition with space for labels	When you want a cleaner pie chart look.
	Funnel Chart	Track sales pipeline drop-off	When visualizing process stages.
	Treemap	Show hierarchy-based share	When comparing large category sets.
	Stacked Column / Bar	Compare parts of a whole by category	When proportions matter with totals visible.
	Waterfall Chart	Show step-by-step change	When visualizing gains/losses over time.

4. Distribution Charts

Purpose: Show data spread, range, and frequency.

Icon	Visual Name	Use Case	When to Use
	Scatter Chart	Compare two numeric variables	When finding relationships or clusters.
	Bubble Chart	Add size to scatter	When showing 3 dimensions of data.
	Histogram	Show frequency distribution	When understanding data spread.
	Box & Whisker (Custom)	Show quartiles & outliers	When exploring statistical data spread.

5. KPI & Summary Visuals

Purpose: Display important performance metrics.

Icon	Visual Name	Use Case	When to Use
	KPI Visual	Show progress toward a goal	When tracking a target value.
	Card	Show a single key metric	When displaying an important number.
	Multi-row Card	Show multiple KPIs in text	When displaying multiple numbers together.
	Gauge Chart	Show percentage toward target	When tracking single-goal progress.
	Smart Narrative	Auto-generate insights	When you want text summaries of visuals.

6. Table-Based Visuals

Purpose: Show detailed structured data.

Icon	Visual Name	Use Case	When to Use
	Table	Show raw data	When showing individual records.
	Matrix	Show pivot table-style data	When summarizing data with dimensions.

7. Geographic / Map Visuals

Purpose: Plot data by location.

Icon	Visual Name	Use Case	When to Use
	Map (Basic)	Show points by location	When plotting coordinates or cities.
	Filled Map (Choropleth)	Show color-coded regions	When comparing values by area.

Icon	Visual Name	Use Case	When to Use
	ArcGIS Maps	Advanced geographic analysis	When adding layers or demographics.
	Shape Map	Show custom boundaries	When plotting on non-standard maps.

8. Advanced / Analytical Visuals

Purpose: Provide deeper analysis & AI-driven insights.

Icon	Visual Name	Use Case	When to Use
	Decomposition Tree	Drill into data step-by-step	When breaking down metrics hierarchically.
	Q&A Visual	Ask questions in natural language	When enabling self-service analytics.
	R Visual	Use R scripts for visuals	When custom statistical visuals are needed.
	Python Visual	Use Python scripts for visuals	When creating advanced ML/AI visuals.

Etl tools

- ❖ **Talend**
- ❖ Apache Nifi
- ❖ **Informatica PowerCenter**
- ❖ **Microsoft SQL Server Integration Services (SSIS)**
- ❖ **Azure Data Factory**
- ❖ AWS Glue
- ❖ Google Cloud Dataflow
- ❖ Apache Airflow
- ❖ Pentaho Data Integration (Kettle)
- ❖ Matillion
- ❖ Fivetran
- ❖ Stitch Data
- ❖ Hevo Data

- ❖ SnapLogic
- ❖ CloverDX
- ❖ IBM DataStage
- ❖ SAP Data Services
- ❖ Oracle Data Integrator (ODI)
- ❖ Alooma
- ❖ Xplenty

Must-know Power Query components:

- ❖ Data Source Connections
- ❖ Navigator Pane
- ❖ Query Editor Window
- ❖ Applied Steps Pane
- ❖ Formula Bar
- ❖ Queries Pane
- ❖ Ribbon Tabs (Home, Transform, Add Column, View, Tools)
- ❖ Data Preview Grid
- ❖ Column Headers
- ❖ Filter and Sort Options
- ❖ Data Type Icons
- ❖ Advanced Editor
- ❖ Parameters
- ❖ Query Dependencies View
- ❖ Merge Queries
- ❖ Append Queries
- ❖ Group By
- ❖ Pivot / Unpivot Columns
- ❖ Custom Column
- ❖ Conditional Column
- ❖ Manage Data Types
- ❖ Replace Values
- ❖ Remove Columns / Rows

- ❖ Keep Columns / Rows
- ❖ Split Column
- ❖ Transform Column (Trim, Clean, Case Change)
- ❖ Fill Up / Down
- ❖ Remove Duplicates
- ❖ Error Handling Tools (Replace Errors, Remove Errors)